UNIVERSITY OF PROFESSIONAL STUDIES, ACCRA

ITIL 4 Foundation

ITIL Management Practices

Chapter 5, Sections 5.2 – 5.2.17.2 | The seventeen service management practices

Lecture deck | 66 slides | Learning objectives, practice-by-practice breakdown, contemporary issues and Ghanaian public-sector applications

Faculty of Information Technology and Communication Studies • Department of Information Technology Studies

SESSION ROADMAP

How this session is organised

1 **Foundations**

What a practice is, how the 34 practices are grouped, and how to read a value-chain heat map.

2 **Practices 5.2.1 – 5.2.4**

Availability, business analysis, capacity and performance, change control.

3 **Practices 5.2.5 – 5.2.9**

Incident, IT asset, monitoring and event, problem, release management.

4 **Practices 5.2.10 – 5.2.13**

Service catalogue, configuration, continuity and service design.

5 **Practices 5.2.14 – 5.2.17**

Service desk, service level, service request, validation and testing.

6 **Beyond the syllabus**

AIOps, FinOps, SRE, platform engineering, and Ghanaian public-sector practice.

BY THE END OF THIS SESSION YOU SHOULD BE ABLE TO

Learning objectives (1 of 2)

1 **Explain the purpose of each practice**

State, in your own words, the purpose of all seventeen service management practices covered in section 5.2.

2 **Recall the examinable definitions**

Define availability, performance, change, incident, problem, known error, workaround, event, IT asset, configuration item, release, service request, service level and service level agreement.

3 **Distinguish closely related concepts**

Separate incident from problem, change control from organisational change management, and service continuity from major incident management.

4 **Map practices to the service value chain**

Identify which value chain activities each practice contributes to most heavily, and justify why.

BY THE END OF THIS SESSION YOU SHOULD BE ABLE TO

Learning objectives (2 of 2)

5 Apply the practices to a real organisation

Given a scenario, recommend which practice or combination of practices addresses the problem described.

6 Evaluate measurement choices

Critique availability percentages, single-system SLA targets and other metrics that look healthy but hide poor customer experience.

7 Relate the practices to Ghanaian institutions

Illustrate each practice using public-sector and financial-sector systems familiar in Ghana, such as GIFMIS, GhIPSS and mobile money platforms.

8 Discuss contemporary developments

Explain how AIOps, cloud FinOps, site reliability engineering and platform engineering are reshaping these practices.

ORIENTATION

Where section 5.2 sits in the ITIL 4 architecture

RECALL FROM CHAPTER 4

The service value system converts demand into value. Inside it, the service value chain provides six activities — plan, improve, engage, design and transition, obtain/build, deliver and support. Practices are the organisational resources that the value chain draws on to actually do the work.

1

Guiding principles

Seven recommendations that shape every decision

2

Governance

Direction, evaluation and oversight by the governing body

3

Service value chain

Six interconnected activities that create the service

4

Practices

34 sets of organisational resources — our focus today

5

Continual improvement

Ongoing refinement across the whole system

GH

A practice is not a process. It bundles people, workflows, tools, information and supplier arrangements — the same four dimensions you met in chapter 3.

CATEGORIES

The thirty-four ITIL practices at a glance

14

General management

Adapted from general business management. Includes strategy management, portfolio management, architecture management, risk management, information security management, knowledge management, measurement and reporting, project management, relationship management, supplier management, workforce and talent management, continual improvement, service financial management and organisational change management.

17

Service management

Developed within service management and IT service management. These are the seventeen practices covered in section 5.2 and examined in detail in this session.

3

Technical management

Adapted from technology management for service management purposes: deployment management, infrastructure and platform management, and software development and management.

GH

Examination note: you are expected to recall the purpose of all 34, but only a subset in detail. Section 5.2 gives you the detailed seventeen.

OUR MAP FOR THE SESSION

The seventeen service management practices

- 5.1 Availability management
- 5.2 Business analysis
- 5.3 Capacity and performance management
- 5.4 Change control
- 5.5 Incident management
- 5.6 IT asset management
- 5.7 Monitoring and event management
- 5.8 Problem management
- 5.9 Release management
- 5.10 Service catalogue management
- 5.11 Service configuration management
- 5.12 Service continuity management
- 5.13 Service design
- 5.14 Service desk
- 5.15 Service level management
- 5.16 Service request management
- 5.17 Service validation and testing

Numbering shown is 5.2.1 to 5.2.17 in the manual, abbreviated here for space.

METHOD

How to read a value-chain heat map

Each practice in the manual is followed by a heat map showing how strongly it contributes to each of the six value chain activities. The shading runs on a four-point scale.

0	1	2	3
No contribution	Low	Medium	High

What the shading is telling you

- Where the practice does most of its work day to day
- Which teams therefore need to own or resource it
- Which value chain activity suffers first when the practice is weak

What it is not telling you

- It is not a maturity score for your organisation
- It is not a priority ranking between practices
- It is illustrative — your own context may differ legitimately

PART ONE

Availability, analysis, capacity and change

- 5.2.1 Availability management
- 5.2.2 Business analysis
- 5.2.3 Capacity and performance management
- 5.2.4 Change control

5.2.1

Availability management

KEY MESSAGE

The purpose of the practice is to make sure services deliver the levels of availability that have been agreed, so that the needs of customers and users are actually met.

DEFINITION — AVAILABILITY

The capability of an IT service, or any other configuration item, to carry out its agreed function at the moment it is needed.

Note the wording

Agreed function, when required. Availability is judged against an agreement and against a moment of need, not against a raw uptime figure.

Core activities of the practice

- Negotiating availability targets that the organisation can realistically achieve
- Designing infrastructure and applications capable of delivering those levels
- Ensuring services and components can generate the data needed to measure availability
- Monitoring, analysing and reporting on availability, and planning improvements to it

5.2.1

Two ways to think about failure: MTBF and MTRS

Availability depends on two things — how often a service fails, and how quickly it comes back.

Mean time between failures (MTBF)

- Measures how frequently the service fails
- A service with an MTBF of four weeks fails roughly thirteen times a year
- Raising MTBF means building components that break less often
- Still the priority where even a brief outage is catastrophic

Mean time to restore service (MTRS)

- Measures how quickly service is restored after a failure
- An MTRS of four hours is an average across many incidents, not a guarantee
- Lowering MTRS means designing for fast, often automatic, recovery
- The dominant modern emphasis in service design

The design shift

Older services were engineered for very high MTBF. The modern emphasis is on minimising MTRS through anti-fragile design — solutions that recover automatically and almost instantly, with negligible business impact.

5.2.1

Measuring availability in ways customers recognise

Percentage availability calculated from MTBF and MTRS rarely matches what customers actually experienced. Four alternatives work better for particular service types.

1 User outage minutes

Incident duration multiplied by users affected, or the sum of minutes lost per user. Suits services that support productivity directly, such as email.

2 Number of lost transactions

Expected transactions in the period minus those that actually completed. Suits transaction-driven processes such as manufacturing or payments.

3 Lost business value

The productivity cost of the failures. Easy for customers to grasp and useful for investment cases, but hard to attribute cleanly to IT causes.

4 User satisfaction

Availability is among the most visible service characteristics, so satisfaction must be checked alongside formally agreed targets.

GH Applied locally: for a mobile money platform, lost transactions during peak market hours communicates far more to a business sponsor than a 99.6 per cent uptime figure.

5.2.1

Who does availability management, and where it lands

Organisational reality

- Most organisations have no dedicated availability management staff
- Activities are distributed — sometimes folded into risk management
- Elsewhere merged with continuity, or with capacity and performance
- Some organisations appoint site reliability engineers for specific products
- Regular testing of failover and recovery mechanisms is essential

Culture beats procedure

Many organisations have a process for calculating and reporting availability metrics. But the manual is blunt about this: availability management is driven as much by culture, experience and knowledge as by following documented procedures. A perfectly written procedure will not save a team that has never rehearsed a failover.

Contribution to the service value chain — involved in all six activities

- Plan: shapes service portfolio decisions and target setting • Improve: ensures improvements do not degrade what already works
- Engage: captures availability requirements • Design and transition: designs to target and tests availability controls
- Obtain/build: a criterion when building or sourcing components • Deliver and support: measurement and event response

5.2.2

Business analysis

KEY MESSAGE

The purpose is to analyse a business, or an element of it, define the needs that emerge, and recommend solutions that address those needs or solve a business problem — in a way that facilitates value creation for stakeholders. It lets an organisation express what it needs, articulate why change is justified, and describe solutions aligned to its objectives.

Key activities

- Analysing systems, processes, services and architectures in a shifting context
- Identifying and prioritising parts of the SVS needing improvement or innovation
- Proposing actions — not only system changes, but process, structural and staffing changes
- Documenting business requirements for supporting services
- Recommending solutions and validating them with stakeholders

Scope beyond software

In IT, business analysis is applied most often to software development projects. But the manual warns explicitly against that restriction: confining business analysis to software development risks producing incomplete solutions. It applies equally to higher-level architectures, to services, and to the service value system as a whole.

The work is performed primarily by business analysts, though others contribute.

5.2.2

Utility and warranty requirements

Business requirements fall into two families. The distinction mirrors utility and warranty from chapter 2.

DEFINITION — UTILITY REQUIREMENTS

Functional requirements that the customer has defined and that are unique to a specific product. What the service must do.

DEFINITION — WARRANTY REQUIREMENTS

Typically non-functional requirements captured from key stakeholders and other practices. Organisations should maintain a library of pre-defined warranty acceptance criteria for reuse.

The analysis trap the manual warns against

- Do not analyse without any intention of acting on the analysis
- Do not analyse so deeply, or for so long, that a timely solution becomes impossible
- Do not try to solve every problem with one massive initiative that delivers too late to be useful

5.2.2

What business analysis needs to succeed

A **Complete access to information**

Analysts need process flows, procedures, work instructions, policies and process metrics — and interviews not only with the process owner but with everyone who participates in each part of it.

B **A recognised skill set**

Critical thinking and evaluation, plus listening, communication and facilitation, the ability to document processes and use cases, and competence in data analysis and modelling.

C **Partner relationships across boundaries**

Where the system crosses organisational boundaries, the units involved must act as partners. Where compromises are needed, a collaborative relationship produces a solution that delivers value to all parties.

D **Interfaces with other practices**

Strategy management sets high-level direction; relationship management secures participation; service design, software development, and measurement and reporting all interact with the analysis.

5.2.3

Capacity and performance management

KEY MESSAGE

The purpose is to ensure that services achieve the performance that has been agreed and expected, satisfying both current and future demand, and doing so cost-effectively.

DEFINITION — PERFORMANCE

A measure of what is achieved or delivered by a system, person, team, practice or service.

Performance rests on capacity

Service capacity is the maximum throughput a configuration item or service can deliver. Specific metrics depend on the technology and the business nature of the service.

Analysis

- Research and monitoring of current service performance
- Capacity and performance modelling

Planning

- Capacity requirements analysis
- Demand forecasting and resource planning
- Performance improvement planning

5.2.3

Capacity and performance across the value chain

Plan

Supports tactical and operational planning with data on actual demand and performance, plus modelling and forecasting tools.

Improve

Improvements are identified and driven by the performance information this practice generates.

Engage

Customer and user expectations are managed using information about capacity constraints and capabilities.

Design and transition

Essential to product and service design — ensures new services are designed for optimum performance, capacity and scalability.

Obtain/build

Ensures components and services being built or obtained meet the organisation's performance needs.

Deliver and support

Services are supported and tested against performance and capacity targets, metrics and reporting tools.

GH

Applied locally: student portal load at UPSA is not uniform. Registration windows and results release produce demand spikes that ordinary term-time capacity planning will miss entirely.

5.2.4

Change control

KEY MESSAGE

The purpose is to maximise the number of successful service and product changes, by ensuring risks are properly assessed, authorising changes to proceed, and managing the change schedule.

DEFINITION — CHANGE

The addition, modification or removal of anything that could have a direct or indirect effect on services.

Scope is set by the organisation

Typically all IT infrastructure, applications, documentation, processes, supplier relationships — and anything else that might affect a product or service directly or indirectly.

The balance the practice must strike

Beneficial changes deliver additional value; the same changes can harm customers and users. Every change should therefore be assessed by people who genuinely understand both the risks and the expected benefits, and must be authorised before deployment. Crucially, that assessment must not introduce unnecessary delay.

5.2.4

Three types of change, managed three different ways

Standard changes

Low-risk, pre-authorised, well understood and fully documented. Implemented without additional authorisation, often initiated as service requests. The risk assessment is done once, when the procedure is created or modified — not on every repetition.

Normal changes

Scheduled, assessed and authorised following a process. Change models determine the roles involved. Low-risk normal changes go to someone who can decide rapidly; very major ones may reach the management board. Triggered by a change request, which CI/CD pipelines increasingly automate.

Emergency changes

Must be implemented as fast as possible — to resolve an incident or apply a security patch. Not normally in the change schedule; assessment and authorisation are expedited. Some documentation may be deferred, and a separate change authority of senior managers often applies.

Examination tip: the distinguishing feature of a standard change is pre-authorisation, not low risk alone.

5.2.4

Change authority and the change schedule

DEFINITION — CHANGE AUTHORITY

The person or group who authorises a change. Assigning the correct authority to each type of change is what makes change control both efficient and effective.

In high-velocity organisations

Decentralising change approval is common practice. Peer review then becomes one of the strongest predictors of high performance.

What the change schedule does

- Helps plan changes and assign resources
- Assists communication and avoids conflicting changes
- After deployment, informs incident and problem management
- Feeds improvement planning

Change control is not organisational change management

Organisational change management handles the people side of change, so that improvements and transformation initiatives actually take hold. Change control is focused on changes to products and services. Confusing the two is a frequent examination error.

PART TWO

Operating the service: incidents, assets, events, problems and releases

- 5.2.5 Incident management
- 5.2.6 IT asset management
- 5.2.7 Monitoring and event management
- 5.2.8 Problem management
- 5.2.9 Release management

5.2.5

Incident management

KEY MESSAGE

The purpose is to minimise the negative impact of incidents by restoring normal service operation as quickly as possible.

DEFINITION — INCIDENT

An unplanned interruption to a service, or a reduction in the quality of a service.

Why it matters disproportionately

Incident management shapes customer and user satisfaction, and how the service provider itself is perceived, more visibly than almost any other practice.

Non-negotiables

- Every incident is logged and managed so it is resolved within the time the customer and user expect
- Target resolution times are agreed, documented and communicated, so expectations stay realistic
- Incidents are prioritised using an agreed classification, so the highest business impact is resolved first
- Separate processes usually exist for major incidents and for information security incidents

5.2.5

Who resolves an incident depends on its complexity

1 The user, via self-help

Use of specific self-help records should be captured for measurement and improvement.

2 The service desk

A substantial share of incidents is resolved at first contact.

3 A support team

More complex incidents are escalated, with routing usually based on incident category.

4 Suppliers or partners

Escalation to third parties who support their own products and services.

5 A temporary swarm team

For the most complex and all major incidents — drawing in provider, suppliers and users together.

6 Disaster recovery plans

In extreme cases, invoked to resolve the incident. Covered by service continuity management, 5.2.12.

GH All of these groups must understand the incident management process and how their contribution manages the value, outcomes, costs and risks of the service.

5.2.5

Swarming, incident records and supplier support

DEFINITION — SWARMING

A technique in which many different stakeholders work together initially, until it becomes clear which of them is best placed to continue and which can move on to other tasks.

Incident records and tooling

- Records stored in a suitable tool, linked to related CIs, changes, problems and known errors
- Modern ITSM tools match incidents automatically to prior incidents, problems or known errors
- Some now generate recommendations for handling future incidents
- Good collaboration tooling matters as much as good workflow tooling

Quality of updates

Updates should cover symptoms, business impact, configuration items affected, actions completed and actions planned — each timestamped and attributed, so everyone involved or interested stays informed.

Third parties in the chain

Third-party components require support agreements that align supplier obligations with the commitments the provider has made to its customers. Routine management of that part of the supplier contract often sits inside incident management. A supplier can even act as the service desk itself, logging and escalating on the provider's behalf.

5.2.6

IT asset management

KEY MESSAGE

The purpose is to plan and manage the full lifecycle of all IT assets, so the organisation can maximise value, control costs, manage risks, support decisions about purchase, reuse, retirement and disposal, and meet regulatory and contractual requirements.

DEFINITION — IT ASSET

Any financially valuable component that can contribute to the delivery of an IT product or service.

DEFINITION — ASSET REGISTER

The accurate inventory information the practice depends on. Best populated by the processes that change asset status, not by periodic audit alone.

Scope

All software, hardware, networking, cloud services and client devices. Sometimes non-IT assets such as buildings or information, where these carry financial value and are required to deliver an IT service. Increasingly it also includes operational technology, including Internet of Things devices that were never traditionally regarded as IT assets but now carry embedded computing and network connectivity.

5.2.6

Asset management, ITAM and SAM

Three nested scopes. Students routinely use the terms interchangeably; the examination does not.

1 Asset management

A well-established practice covering the acquisition, operation, care and disposal of organisational assets, particularly critical infrastructure.

2 IT asset management (ITAM)

A sub-practice of asset management, aimed specifically at managing the lifecycles and total costs of IT equipment and infrastructure.

3 Software asset management (SAM)

An aspect of ITAM, aimed at the acquisition, development, release, deployment, maintenance and eventual retirement of software assets.

Software runs on hardware, so software and hardware assets should be managed together to keep licences properly controlled — and cloud-based assets belong in the same regime.

5.2.6

Requirements differ sharply by asset type

H Hardware assets

Must be labelled for identification. Location must be known, and assets protected from theft, damage and data leakage. Reuse or decommissioning may need special handling — disk erasure or shredding, depending on security requirements. Regulatory duties such as electrical waste directives may apply.

S Software assets

Must be protected from unlawful copying that would create unlicensed use. Licence terms must be adhered to and reuse kept within the contract. Verified proof of purchase and entitlement must be retained. Licences are easily lost at decommissioning, so the process must recover and re-pool them.

C Cloud-based assets

Must be assigned to specific products or groups so costs can be tracked. Funding must permit new instances when needed and removal when not — without uncontrolled cost. Contractual arrangements are handled as strictly as software licences.

D Client assets

Must be assigned to named individuals who take responsibility for their care. Processes are needed for lost or stolen devices, together with tooling to erase sensitive data remotely or otherwise prevent its loss with the device.

GH Cost control warning: individual cloud usage looks cheap, but small amounts accumulate. Without ITAM discipline organisations consume far more than planned and receive a correspondingly large bill.

5.2.6

Asset register, CMS federation and organisation

Federating with the CMS

The asset register holds information about assets, their costs and related contracts, so it is often combined or federated with the configuration management system. Where the two stay separate, assets must be mappable between them — usually through a standard naming convention. Federation with financial and supplier systems may also be necessary.

Centralised or distributed?

Some organisations run a centralised ITAM team, which may also own configuration management. Others make each technical team responsible for the assets it supports — storage team for storage, network team for network. Context and culture decide. But some central roles help ensure data quality and build expertise in licensing and inventory systems.

Typical activities

- Define, populate and maintain the asset register — structure, content, and storage facilities for assets and related media
- Control the asset lifecycle with other practices — upgrading obsolete software, onboarding staff with a laptop and phone — and record every status, location, characteristic and assignment change
- Provide current and historical data, reports and support about IT assets to other practices
- Audit assets, media and conformity with regulations and licence terms, and drive corrective and preventive improvement

5.2.7

Monitoring and event management

KEY MESSAGE

The purpose is to observe services and service components systematically, and to record and report selected changes of state identified as events. The practice identifies and prioritises infrastructure, service, business process and information security events, and establishes the appropriate response — including responses to conditions that could lead to faults or incidents.

DEFINITION — EVENT

Any change of state that has significance for the management of a service or other configuration item. Events are typically recognised through notifications created by an IT service, a CI, or a monitoring tool.

The monitoring half

- Systematic observation of services and underpinning CIs
- Detects conditions of potential significance
- Should be highly automated; active or passive

The event management half

- Records and manages state changes defined as events
- Determines significance and initiates the control action
- Not all monitoring yields an event

5.2.7

Three classes of event, three different responses

Informational

No action required at the time of identification. Value comes later: analysing the accumulated data may reveal desirable proactive steps that benefit the service.

Warning

Allows action to be taken before any negative impact is actually experienced by the business. This is the pre-emptive band, and the reason monitoring earns its cost.

Exception

A breach of an established norm has been identified — for example, against a service level agreement. Action is required even though business impact may not yet have been experienced.

Key activities

- Decide what should be monitored and set the monitoring strategy
- Implement and maintain monitoring — native features plus purpose-built tools
- Establish thresholds and criteria defining each event type
- Set policies for handling each type, then automate them

Active versus passive monitoring

In active monitoring, tools poll key configuration items and generate alerts when an exception condition appears. In passive monitoring, the configuration item itself generates the operational alert. Automated correlation of events matters as much as detection: volumes are enormous, and without policies on how to limit, filter and use the data it carries no value.

5.2.8

Problem management

KEY MESSAGE

The purpose is to reduce the likelihood and impact of incidents, by identifying actual and potential causes of incidents and managing workarounds and known errors.

DEFINITION — PROBLEM

A cause, or a potential cause, of one or more incidents.

DEFINITION — KNOWN ERROR

A problem that has been analysed but has not been resolved.

DEFINITION — WORKAROUND

A solution reducing or eliminating the impact of an incident or problem for which full resolution is not yet available.

Incidents

- Have an impact on users or business processes
- Must be resolved so normal business activity resumes
- The clock is the customer's

Problems

- Are the causes of incidents
- Require investigation and analysis to identify causes
- Produce workarounds and longer-term resolutions, reducing future incidents

5.2.8

The three phases of problem management

1

Problem identification

Identify and log problems: trend analysis of incident records, duplicate and recurring issues spotted by users, desk and technical staff, recurrence risk noted during major incidents, plus information from suppliers, partners, developers and test teams.

2

Problem control

Analyse problems and document workarounds and known errors. Problems are prioritised by risk and managed as risks, on impact and probability. Not every problem must be analysed — progress on the highest-priority ones beats investigating every minor one.

3

Error control

Manage known errors where initial analysis is complete and faulty components identified. Identify potential permanent solutions, which may trigger a change request — but only where justified on cost, risk and benefit.

Analyse across all four dimensions

Incidents typically have many interrelated causes. Problem control must consider every contributory cause, including those that lengthened duration or worsened impact. An incident caused by inaccurate documentation may require correcting the document and training support staff, suppliers and users.

Living with workarounds

An effective workaround can become the permanent answer where resolution is not viable or cost-effective. The problem then remains a known error, and the documented workaround is applied when related incidents occur. Every workaround must define the symptoms it applies to — and its effectiveness should be re-evaluated each time it is used.

5.2.8

Where problem management touches other practices

A **Risk management**

Problem management can be organised as a specific case of risk management — identifying, assessing and controlling risks across all four dimensions. Adopting risk tools and techniques is useful.

B **Change control**

Implementing a resolution usually sits outside its scope. It initiates resolution via change control and joins the post-implementation review, but approving and implementing changes is not its responsibility.

C **Knowledge management**

Output includes documentation of workarounds and known errors; the practice also draws on the knowledge management system to investigate, diagnose and resolve.

D **Continual improvement**

Improvement opportunities across all four dimensions may enter the continual improvement register, prioritised with continual improvement techniques and sometimes managed in a product backlog.

GH Complement or conflict: identifying a cause may resolve an incident faster — or investigating a cause may delay the actions needed to restore service. The two must be designed to work together.

5.2.9

Release management

KEY MESSAGE

The purpose is to make new and changed services and features available for use.

DEFINITION — RELEASE

A version of a service or other configuration item, or a collection of configuration items, that is made available for use.

What a release contains

Infrastructure and application components working together, plus documentation, training for users or IT staff, updated processes or tools — developed in house or procured and integrated.

Scale and planning

- From one minor changed feature to a completely new service
- A release plan specifies the exact combination of components and the timing
- A release schedule documents timing, negotiated and agreed with customers
- A post-implementation review enables learning and confirms customer satisfaction

Staged releases

Release is often staged: a pilot goes to a small group to confirm everything works before wider groups receive it. Sometimes staging is impossible — where a major restructuring of shared underlying data is required, everyone must receive the release simultaneously.

5.2.9

Release in waterfall and in Agile/DevOps environments

Traditional / waterfall

- Almost all release work happens before deployment
- Plans specify exactly which components deploy in the release
- Deployment then makes the new functionality available
- Release management and deployment may be combined into a single process

Agile / DevOps

- Significant release activity happens after deployment
- Software and infrastructure deploy in many small increments
- Release activity enables the new functionality later — often as a very small change
- Typically integrated with the continuous integration and delivery toolchain

Blue/green releases

Two mirrored production environments. Users are switched to the environment carrying the new functionality by network tools that connect them to the correct one.

Feature flags

Specific features are released to individual users or groups in a controlled way. The functionality is deployed to production without being released; a user configuration setting then switches it on for chosen users.

PART THREE

Describing, configuring, protecting and designing the service

- 5.2.10 Service catalogue management
- 5.2.11 Service configuration management
- 5.2.12 Service continuity management
- 5.2.13 Service design

5.2.10

Service catalogue management

KEY MESSAGE

The purpose is to provide a single source of consistent information on all services and service offerings, and to make sure it reaches the relevant audience.

Catalogue versus portfolio

The catalogue lists services currently available. It is a subset of the fuller list tracked in the provider's service portfolio. The catalogue can be a document, an online portal, or a tool.

Ongoing activities

Publishing, editing and maintaining service and product descriptions and their offerings. It shows what is available, and on what terms. Supported by roles such as the service owner as services are introduced, changed or retired.

1 User views

Information on service offerings that can be requested, plus provisioning details.

2 Customer views

Service level, financial and service performance data.

3 IT to IT customer views

Technical, security and process information for use in service delivery.

5.2.10

From static list to engagement tool

The adoption problem

For the catalogue to be seen as useful by the customer organisation it must do more than publish information statically. Unless it enables customer engagement — supporting discussion of standard and non-standard offerings, or automating request and order fulfilment — its chances of ongoing adoption as a meaningful resource are minimal.

DEFINITION — REQUEST CATALOGUE

A view of the service catalogue providing details on service requests for existing and new services, made available to the user.

Why organisations gravitate to it

Because the request catalogue focuses on the consumable, orderable elements of service offerings — the part users can act on directly.

GH

Applied locally: an intranet page listing IT services is not a service catalogue in this sense. Until a lecturer can request a projector replacement or a lab account from it, it remains a notice board.

Value chain: involved in all six activities, most visibly engage — enabling and partly automating relationship management, request management and the service desk.

5.2.11

Service configuration management

KEY MESSAGE

The purpose is to ensure that accurate, reliable information about the configuration of services and the CIs supporting them is available when and where it is needed — including how CIs are configured and the relationships between them.

DEFINITION — CONFIGURATION ITEM (CI)

Any component that needs to be managed in order to deliver an IT service.

DEFINITION — CONFIGURATION MANAGEMENT SYSTEM (CMS)

A set of tools, data and information used to support service configuration management.

What is collected, and why relationships matter more than counts

CIs typically include hardware, software, networks, buildings, people, suppliers and documentation — and services themselves are treated as CIs. The practice explains how the many CIs contributing to a service work together: how they interact, relate and depend on each other to create value, including dependencies between services. That high-level view is often called a service map or service model, and forms part of the service architecture.

5.2.11

Balancing the cost of configuration data against its value

The discipline the manual insists on

Maintaining large amounts of detailed information about every component and its relationships can be costly and may deliver very little value. Requirements must be based on the organisation's goals and on how configuration management actually contributes to value creation.

Plan by asking four questions

- Who needs the configuration information?
- How will it be used?
- What is the best way for them to obtain it?
- Who can maintain and update it?

CMDB, federation and sensitivity

- Information may sit in one CMDB, but is more commonly distributed across sources
- Links between configuration records must be maintained either way
- Some organisations federate CMDBs for an integrated view
- Some information is sensitive — useful to an attacker, or personal data about users

Processes typically needed

- Identify new CIs and add them to the CMS
- Update configuration data when changes are deployed
- Verify that configuration records are correct
- Audit applications and infrastructure to find anything undocumented

5.2.12

Service continuity management

KEY MESSAGE

The purpose is to ensure that the availability and performance of a service are maintained at sufficient levels in the event of a disaster. The practice provides a framework for organisational resilience, producing a response that safeguards key stakeholders and the organisation's reputation, brand and value-creating activities.

Trigger threshold

It engages when a service disruption or organisational risk exceeds what normal response and recovery practices — including major incident management — can absorb. An event of that magnitude is termed a disaster.

Defining a disaster in advance

Each organisation must define what counts as a disaster in its own context, at both organisational and per-service level, using a business impact analysis — and it must do so before a trigger event, not during one.

Disaster sources the manual lists

Supply chain failure • Terrorism • Weather • Cyber attack • Health emergency • Political or economic event • Technology failure • Public crisis
Organisational impact ranges from lost income and damaged reputation to breach of law and safety regulations, and risk to personal safety.

5.2.12

The four examinable continuity terms

RTO — Recovery time objective

The maximum acceptable period following a disruption that can elapse before the absence of business functionality severely impacts the organisation — the maximum agreed time within which a product or activity must be resumed, or resources recovered.

RPO — Recovery point objective

The point to which information used by an activity must be restored, so the activity can operate on resumption. In practice, how much data loss is tolerable.

DRP — Disaster recovery plans

A set of clearly defined plans for how the organisation will recover from a disaster and return to a pre-disaster condition, considering all four dimensions of service management.

BIA — Business impact analysis

Identifies vital business functions and their dependencies — suppliers, people, other processes, IT services. It defines recovery requirements: RTOs, RPOs and minimum target service levels for each IT service.

GH Distinguish carefully: RTO is about time to restore the service; RPO is about how far back the data must be recoverable. Examinations test the pair together.

5.2.12

Service continuity or major incident? Draw the line first

Service continuity management

- Focuses on events the business considers significant enough to be a disaster
- Invokes pre-defined disaster recovery plans
- Concerned with organisational survival and resilience
- Must be thought through, designed for flexibility and tested regularly

Incident and major incident management

- Handles less significant events within normal response capability
- Restores normal service operation as quickly as possible
- Concerned with day-to-day service quality
- Escalates upward when thresholds are breached

The distinction must be documented before the event

The manual is explicit: the distinction between disasters, major incidents and incidents needs to be pre-defined, agreed and documented, with clear thresholds and triggers for calling the next tier of response and recovery into action — without unnecessary delay and risk.

High availability itself comes from a combination: business planning, technical architecture resilience, availability planning, proactive risk and information security management, and incident and problem management working together.

5.2.13

Service design

KEY MESSAGE

The purpose is to design products and services that are fit for purpose, fit for use, and deliverable by the organisation and its ecosystem. This includes planning and organising people, partners and suppliers, information, communication, technology and practices — and the interaction between the organisation and its customers.

What happens without it

- Products and services become unduly expensive to run and prone to failure
- Resources are wasted; the service is neither customer-centred nor holistically designed
- No improvement programme is likely to achieve what proper design would have achieved first time
- Cost-effective services delivering what customers expect become extremely hard to achieve

Design decisions never sit in isolation

- Impact on other products and services
- All relevant parties, including customers and suppliers
- The existing architectures and the required technology
- The service management practices themselves
- The necessary measurements and metrics
- And retirement — an unplanned withdrawal harms customers avoidably

5.2.13.1

Design thinking

A practical, human-centred approach that accelerates innovation — complementary to Lean and Agile, drawing on logic, imagination, intuition and systems thinking.

1

Inspiration and empathy

Direct observation of people and how they work with products and services, and how they might interact differently.

2

Ideation

Divergent thinking generates many unique ideas; convergent thinking narrows them to a preferred solution.

3

Prototyping

Ideas tested early, iterated and refined. Prototypes gather feedback and expose strengths and weaknesses fast.

4

Implementation

Concepts brought to life, coordinated with all relevant practices. Agile can deliver the solution iteratively.

5

Evaluation

With project and release management, measures actual performance against acceptance criteria and finds improvement opportunities.

Why multi-disciplinary teams

Design thinking balances the perspectives of customers, technology, the organisation, partners and suppliers. That makes it highly integrative, well aligned to the service value system, and a key enabler of digital transformation.

5.2.13.2

Customer experience, user experience and Lean UX

Customer experience (CX)

- Manages every aspect of the complete experience
- Covers time, quality, cost, reliability and effectiveness
- The whole journey from demand to value realisation

User experience (UX)

- Looks specifically at ease of use of the product or service
- Concerns how the customer interacts with it
- Narrower, more immediate, more testable

Lean UX

A mindset, a culture and a process embracing Lean–Agile methods. It implements functionality in minimum viable increments and judges success by measuring results against an outcome hypothesis. The core objective is to obtain feedback as early as possible so that quick decisions can be made.

Typical Lean UX questions

- Who are the customers of this product or service, and what will it be used for?
- When is it used, and under what circumstances?
- What will be the most important functionality?
- What are the biggest risks?
- Assumptions are then prioritised by the risk they represent

PART FOUR

Meeting the user: desk, service levels, requests and testing

- 5.2.14 Service desk
- 5.2.15 Service level management
- 5.2.16 Service request management
- 5.2.17 Service validation and testing

5.2.14

Service desk

KEY MESSAGE

The purpose is to capture demand for incident resolution and service requests. It should also be the entry point and single point of contact between the service provider and all of its users.

The model varies; the function does not

Delivery may run from a physical team working shifts, to a distributed mix of people connected virtually, to automated technology and bots. The function and the value remain the same regardless of the model.

The shift in what desks do

With increased automation and the gradual removal of technical debt, the focus has moved to supporting people and business rather than technical issues alone. Desks increasingly arrange, explain and coordinate rather than only fix broken technology.

Two points the manual stresses

- However efficient the desk, some issues will always need escalation and underpinning support — support and development teams must collaborate closely to present a joined-up approach
- The desk should be the empathetic and informed link between provider and users, adding value by understanding and acting on the business context of an action, not merely logging it

5.2.14

Channels, technology and skills

Access channels

- Phone, including IVR, conference calls and voice recognition
- Service portals and mobile applications, backed by catalogues and knowledge bases
- Live chat and chatbots
- Email for logging, updating and follow-up surveys
- Walk-in desks, increasingly common in higher education
- Text and social media messaging; public and corporate forums for peer support

Supporting technology for a centralised desk

- Intelligent telephony — computer-telephony integration, IVR, automatic call distribution
- Workflow systems for routing and escalation
- Workforce management and resource planning systems
- A knowledge base; call recording and quality control
- Remote access tools; dashboards and monitoring
- Configuration management systems

Virtual desks

Agents work from multiple geographically dispersed locations. This requires more sophisticated technology, with more complex routing and escalation — solutions are often cloud-based.

Skills that matter most

Empathy, incident analysis and prioritisation, effective communication and emotional intelligence. The key skill is understanding and diagnosing an incident in terms of business priority, then acting to resolve it with the skills, knowledge, people and processes available.

5.2.15

Service level management

KEY MESSAGE

The purpose is to set clear, business-based targets for service levels, and to ensure that service delivery is properly assessed, monitored and managed against those targets.

DEFINITION — SERVICE LEVEL

One or more metrics that define expected or achieved service quality.

DEFINITION — SERVICE LEVEL AGREEMENT (SLA)

A documented agreement between a service provider and a customer identifying both the services required and the expected level of service.

Service level management provides end-to-end visibility. To achieve it, the practice:

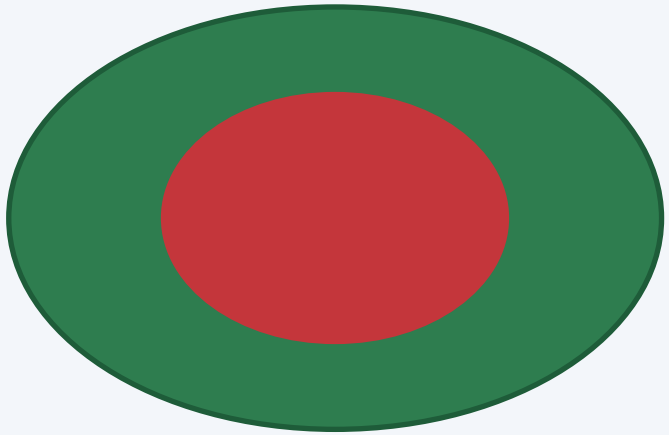
- Establishes a shared view of services and target service levels with customers
- Ensures defined levels are met, through collection, analysis, storage and reporting of relevant metrics
- Performs service reviews, confirming the current set of services still meets organisational and customer needs
- Captures and reports service issues, including performance against defined service levels

5.2.15

The watermelon SLA effect

Green outside, red inside

Traditional SLAs were built on individual activities: incident resolution times, system availability figures, volume counts. Without business context these are often meaningless. A system availability of 99.6 per cent looks impressive — but the acceptable 0.4 per cent of unavailability becomes intolerable if it falls during a commercial transaction, an operating theatre in use, or point-of-sale tills in use. Satisfaction will be low regardless of whether the SLA was met.



SLA reports

Requirements for a successful SLA

- Related to a defined service in the catalogue, not a free-floating metric
- Related to defined outcomes, using balanced bundles of metrics
- A genuine agreement, discussed with partners, sponsors, users and customers
- Simply written, easy to understand and use for all parties

The corrective

Service level management identifies metrics that truthfully reflect the customer's actual experience and satisfaction with the whole service. These vary across organisations, and the only way to learn what they are is to find out directly from customers.

5.2.15

What service level management listens to

1 Customer engagement

Initial listening, discovery and information capture. Open questions work best: what differentiates a good day from a bad one, and what is the best measure of your success?

2 Customer feedback

Gathered formally and informally. Surveys range from immediate follow-up questions after an incident to reflective periodic surveys on overall experience — both are event-based.

3 Key business-related measures

Agreed on what the customer values: a bundle of SLA metrics, or a very specific activity such as a sales transaction, a project completion, or getting an ambulance to an accident within a stated time.

4 Operational and business metrics

Operational metrics are low-level indicators — availability, response and fix times, request processing times. Business metrics are any activity the customer values, such as passenger check-in.

5.2.16

Service request management

KEY MESSAGE

The purpose is to support the agreed quality of a service by handling all pre-defined, user-initiated service requests in an effective and user-friendly manner.

DEFINITION — SERVICE REQUEST

A request from a user, or a user's authorised representative, that initiates a service action agreed as a normal part of service delivery.

A service delivery action

Providing a report, replacing a toner cartridge

Information

How to create a document, or what the office hours are

Provision of a resource

A phone or laptop for a user, a virtual server for a development team

Access to a resource

Access to a particular file or folder

Feedback and complaints

Complaints about a new interface, or compliments to a support team

5.2.16

Making service requests work at scale

Why standardisation is possible here and not elsewhere

Because requests are pre-defined and pre-agreed as a normal part of service delivery, they can be formalised with a clear standard procedure for initiation, approval, fulfilment and management. Some have very simple workflows — a request for information. Others, such as setting up a new employee, are complex and draw on many teams and systems. Either way, the steps must be well known and proven so fulfilment times can be agreed and status communicated.

1 Standardise and automate

Standardise and automate requests and their fulfilment as far as possible.

2 Streamline approvals

Set policies on which requests need limited or no extra approval.

3 Set expectations honestly

Base fulfilment-time expectations on what can realistically be delivered.

4 Improve continuously

Find opportunities for faster fulfilment and more automation.

5 Redirect misfiled work

Redirect requests that are really incidents or changes.

6 Reuse workflow models

Few workflow models, reused for new requests, aid maintainability.

5.2.17

Service validation and testing

KEY MESSAGE

The purpose is to ensure that new or changed products and services meet defined requirements. Service value is defined from customer input, business objectives and regulatory requirements, documented within design and transition, and used to establish measurable quality and performance indicators supporting assurance criteria and testing requirements.

Service validation (5.2.17.1)

Establishes deployment and release acceptance criteria — the conditions for production readiness — which are then verified through testing. Criteria may be utility- or warranty-focused, derived from customer, regulatory, business, risk and security requirements.

Test strategy (5.2.17.2)

Defines the overall approach to testing. It can apply to an environment, a platform, a set of services or one service. It applies equally to in-house and externally developed solutions, and must match the stakeholders' risk appetite.

GH

Applied locally: procurement of a vendor-built system does not transfer the testing obligation. The manual is explicit that externally developed solutions are tested on the same footing as in-house builds.

Value chain: involved in every activity except plan — with the heaviest contribution to design and transition, and obtain/build.

5.2.17.2

Typical test types: utility and warranty

Utility / functional tests

- Unit test — a test of a single system component
- System test — overall testing of the system, including software and platforms
- Integration test — testing a group of dependent software modules together
- Regression test — testing whether previously working functions were impacted

Warranty / non-functional tests

- Performance and capacity test — speed and capacity under load
- Security test — vulnerability, policy compliance, penetration, denial of service risk
- Compliance test — legal and regulatory requirements have been met
- Operational test — backup, event monitoring, failover, recovery and reporting
- Warranty requirements test — documentation, training, support model, knowledge transfer
- User acceptance test — performed by users to approve a release

The metrics that matter

Escaped defects, test coverage and service performance against SLAs are critical success measures — they are what allow customer experience to improve and risk to fall.

PART FIVE

Contemporary issues in the field

- AIOps and machine learning in operations
- Cloud economics, FinOps and asset sprawl
- Site reliability engineering and platform engineering
- Cybersecurity, data protection and resilience in Ghana

CONTEMPORARY ISSUE 1

AIOps: machine learning inside the operational practices

The manual already anticipates this — modern ITSM tools automatically match incidents to prior incidents, problems and known errors, and generate recommendations. The direction of travel since has been sharper.

1 Event noise reduction and correlation

Machine learning collapses thousands of correlated alerts into a handful of actionable signals — directly addressing the manual's warning that huge data volumes without filtering policies carry no value.

2 Anomaly detection replacing static thresholds

Rather than fixed warning and exception thresholds, models learn normal behaviour per service and per season. This changes how monitoring and event management sets its criteria.

3 Assisted diagnosis in problem management

Causal inference over incident and change records surfaces candidate causes. Human judgement still decides — the manual insists diagnosis relies on knowledge and experience, not procedure.

4 Conversational first line at the service desk

Large language models handle a growing share of requests and simple incidents. The practice question becomes where the empathetic, informed human link is still required.

CONTEMPORARY ISSUE 2

Cloud economics, FinOps and the sprawl problem

The manual's warning, restated

Each individual use of a cloud service may be relatively cheap, but spending in small amounts makes it easy to consume far more resource than planned — leaving the organisation with a correspondingly large bill. FinOps is the discipline that grew up around exactly this failure.

1 Showback and chargeback

Cloud assets assigned to specific products or groups so cost is visible to the team that generates it — the manual's own requirement, now operationalised.

2 Rightsizing and reservation

Continuous matching of provisioned capacity to observed demand, bringing capacity and performance management directly into the cost conversation.

3 Licence entitlement in the cloud

Bring-your-own-licence models make SAM harder, not easier. Entitlement records must survive migration or organisations pay twice.

4 SaaS discovery

Departments procure SaaS without IT involvement. The asset register cannot record what it never learns about, and audits become the only detection mechanism.

GH Applied locally: foreign-currency cloud billing exposes Ghanaian institutions to exchange-rate movement on top of consumption growth. Budgeting in cedis against a dollar-denominated bill is itself a capacity and financial management risk.

CONTEMPORARY ISSUE 3

SRE, DevOps and platform engineering

Where ITIL 4 already meets SRE

- The manual names site reliability engineers as owners of availability for specific products
- Anti-fragile design and MTRS minimisation are core SRE positions
- Decentralised change approval with peer review as a performance predictor is a DevOps finding
- Blue/green releases and feature flags are described directly in release management

Where the tension remains

- Error budgets set a deliberate tolerance for failure; SLAs traditionally do not
- Toil reduction can conflict with the documentation expectations of change control
- You build it, you run it blurs the service desk's single point of contact
- Platform engineering creates internal developer platforms that behave as services with their own catalogues, SLAs and desks

The synthesis that most organisations arrive at

Practices remain the shared vocabulary and the accountability structure; engineering culture supplies the speed. ITIL 4 was deliberately rewritten to permit this — which is why change control replaced change management, and why the manual repeatedly warns that assessment must not introduce unnecessary delay.

CONTEMPORARY ISSUE 4

Security, data protection and resilience in the Ghanaian context

A Data protection duties inside configuration management

The manual warns that configuration information may include personal data such as phone numbers and home addresses. Under Ghana's Data Protection Act, 2012 (Act 843), that makes the CMDB itself a processing activity requiring lawful basis and safeguards.

B Cybersecurity as a continuity trigger

Cyber attack sits in the manual's list of disaster sources. The Cybersecurity Act, 2020 (Act 1038) and the incident reporting expectations that follow from it turn parts of incident and continuity management into statutory obligations for designated critical information infrastructure.

C Power and connectivity as availability constraints

MTBF and MTRS calculations that assume stable grid supply understate real risk. Availability design in Ghana must treat power continuity and redundant connectivity as first-class components, not facilities issues.

D Public-sector service catalogues and citizen expectation

Digital public services — tax, procurement, identity and payments platforms — are now consumed directly by citizens. Watermelon SLAs are politically costly when the user is a taxpayer rather than an internal colleague.

CONSOLIDATION

Which practices concentrate where in the value chain

Involved in all six activities	Availability management, business analysis, capacity and performance management, change control, release management, service catalogue management, service configuration management, service continuity management, service design
All activities except plan	Monitoring and event management, service desk, service request management, service validation and testing
Concentrated in engage, and deliver and support	Incident management
Concentrated in improve, and deliver and support	Problem management
Concentrated in design and transition, and obtain/build	IT asset management
Concentrated in plan and engage	Service level management

Read as typical contribution, not as a maturity or priority ranking.

KEY TAKEAWAYS

Six things to carry out of this session

1 Practices are resources, not procedures

Each bundles people, information, technology, partners and workflows across all four dimensions.

2 Restoration and diagnosis are different jobs

Incident management restores service. Problem management finds and removes causes. Designed together, they complement; conflated, they conflict.

3 Measurement must reflect experience

Percentage availability and single-system SLA targets can be met while customers are failed. Ask customers what success means.

4 Control must not become delay

Correct change authority, pre-authorised standard changes and decentralised approval with peer review are how control and speed coexist.

5 Know the threshold before the event

Incident, major incident and disaster must be distinguished, agreed and documented in advance, with clear triggers.

6 Design decisions are never isolated

Every change to a service touches other services, parties, architectures, practices and metrics — including its eventual retirement.

ASSESSMENT PREPARATION

Review and discussion questions

Recall — attempt without notes

- Define incident, problem, known error and workaround, and state how each relates to the others.
- Distinguish standard, normal and emergency changes by their authorisation route.
- State the difference between RTO and RPO.
- Which practices are involved in every value chain activity except plan?

Application — 10 marks each

- A university portal is available 99.7 per cent of the year but fails during every registration window. Diagnose the failure using service level management.
- Design a request catalogue for a Ghanaian district assembly's IT unit, naming five request types and their workflows.

Evaluation — seminar debate, 20 marks

- “In an organisation practising continuous deployment, formal change control adds cost without adding safety.” Argue for and against, using evidence from section 5.2.4, and reach a defended position.
- Assess whether the availability measurements in section 5.2.1 are adequate for a mobile money service operating in a market with intermittent power supply. Recommend and justify two additional measures.

REFERENCES

Reading and acknowledgements

Primary source

AXELOS (2019). ITIL® Foundation: ITIL 4 Edition. London: TSO. Chapter 5, sections 5.2 to 5.2.17.2 — ITIL management practices.

All definitions, purpose statements and value chain contributions in this deck are summarised from that chapter. ITIL® is a registered trade mark of AXELOS Limited.

Recommended supplementary reading

- AXELOS, ITIL 4: Create, Deliver and Support (for practice depth)
- Beyer et al., Site Reliability Engineering (for the MTRS argument)
- Forsgren, Humble and Kim, Accelerate (for peer review as a performance predictor)
- Ghana Data Protection Act, 2012 (Act 843); Cybersecurity Act, 2020 (Act 1038)

Preparation for the next session

Read section 5.1, the general management practices, and section 5.3, the technical management practices. Come prepared to argue which single practice from section 5.2 your own organisation performs worst, and what evidence supports that judgement.

University of Professional Studies, Accra • Faculty of Information Technology and Communication Studies